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CHAPTER 3

PUBLIC KEY CRYPTO

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APPENDIX

MODULAR ARITHMETIC

KNAPSACK

RSA

MODULAR ARITHMETIC



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- For integer x and n , “ $x \bmod n$ ” is the **remainder** of x / n .

- Examples

$$7 \bmod 6 = 1$$

$$33 \bmod 5 = 3$$

$$33 \bmod 6 = 3$$

$$51 \bmod 17 = 0$$

$$17 \bmod 6 = 5$$

Practice

$$6 \bmod 5 = ?$$

$$23 \bmod 5 = ?$$

$$10 \bmod 5 = ?$$

$$58 \bmod 20 = ?$$

$$100 \bmod 20 = ?$$

MODULAR ADDITION



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■ Notation and facts

- $7 \bmod 6 = 1$
- $7 = 13 = 1 \bmod 6$
- $((a \bmod n) + (b \bmod n)) \bmod n = (a + b) \bmod n$
- $((a \bmod n)(b \bmod n)) \bmod n = ab \bmod n$

■ Addition example

- $3 + 5 = 2 \bmod 6$
- $2 + 4 = 0 \bmod 6$
- $3 + 3 = 0 \bmod 6$
- $(7 + 12) \bmod 6 = 19 \bmod 6 = 1 \bmod 6$
- $(7 + 12) \bmod 6 = (1 + 0) \bmod 6 = 1 \bmod 6$

■ Multiplication example

- $3 \cdot 4 = 0 \bmod 6$
- $2 \cdot 4 = 2 \bmod 6$
- $5 \cdot 5 = 1 \bmod 6$
- $(7 \cdot 4) \bmod 6 = 28 \bmod 6 = 4 \bmod 6$
- $(7 \cdot 4) \bmod 6 = (1 \cdot 4) \bmod 6 = 4 \bmod 6$

MODULAR INVERSE



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- *Additive inverse* of $x \bmod n$, denoted as $-x \bmod n$, is the number that must be added to x to get $0 \bmod n$.
 - $-2 \bmod 4 = \bar{6}$ since $2+4 = 0+6$
- *Multiplicative inverse* of $x \bmod n$, denoted $x^{-1} \bmod n$, is the number that must be multiplicative by x to get $1 \bmod n$.
 - $3^{-1} \bmod 7 = 5$; since $3 \cdot 5 = 1 \bmod 7$

TOTIENT FUNCTION



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- $\phi(n)$ is the number of numbers less than n that are relatively prime to n .
 - Positive integer.
- Example
 - $\phi(4) = 2$ since 4 is relatively prime to 3, 1.
 - $\phi(5) = 4$ since 5 is relatively prime to 1, 2, 3, 4
 - $\phi(12) = 4$
 - $\phi(p) = p-1$ if p is prime.
 - $\phi(pq) = (p-1)(q-1)$ if p and q prime



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- *Additive inverse* of $x \bmod n$, denoted as $-x \bmod n$, is the number that must be added to x to get $0 \bmod n$.
 - $-2 \bmod 4 = 4$, since $2+4 = 0 \bmod 6$
- *Multiplicative inverse* of $x \bmod n$, denoted $x^{-1} \bmod n$, is the number that must be multiplicative by x to get $1 \bmod n$.
 - $3^{-1} \bmod 7 = 5$; since $3 \cdot 5 = 1 \bmod 7$

$$\mathbb{Z}_{26} = (0, 1, 2, \dots, 25)$$

mod 26

tion

$$11^{-1} \bmod 26 = 19$$

$$\text{Gcd}(273, 301) = 77$$

g	π_1	π_2	π	t_1	t_2	t
0	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
3	3	1	0	5	-7	26

$$\begin{aligned}
 11^{-1} \bmod 26 &= -7 \\
 &= -7 + 26 \\
 &= 19
 \end{aligned}$$

$$t = t_i - g t_{i+1}$$

$$0 - 2 \cdot 1 = -2$$

$$1 - (2 \times -2) = 5$$

$$-2 - (1 \times 5) = -7$$

$$5 - (3 \times -7) = 26$$

$$= 26$$

$$* 17^{-1} \bmod 203$$

$$\text{GCD}(17, 203) = 1$$

$$17 \mid 203 \mid 11 \rightarrow$$

$$\begin{array}{r} 17 \\ 33 \\ 17 \\ \hline 16 \end{array}$$

$$16 \mid 17 \mid 1$$

$$\begin{array}{r} 16 \\ 16 \\ \hline 1 \end{array}$$

$$11 \mid 16 \mid 16$$

$$\begin{array}{r} 16 \\ 16 \\ \hline X \end{array}$$

$$2 \mid 16 \mid 8$$

$$\begin{array}{r} 16 \\ 16 \\ \hline 0 \end{array}$$

	g	π_1	π_2	π	t_1	t_2	t
- 11	203	17	16	0	1	-11	
- 1	17	16	1	1	-11	12	
- 16	16	1	0	-11	12		

$$17^{-1} \bmod 181$$

$$t = t_i - g t_{i+1}$$

$$t = 0 - 11 \times 1$$

$$= -11$$

$$1 - (1 \times -11)$$

$$1 + 11$$

$$= 12$$



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$$11^{-1} \bmod 26 = 19$$

$$17^{-1} \bmod 203 = 12$$

$$28^{-1} \bmod 19 = ?$$

PKC IS NEWCOMER



- Different name
 - Asymmetric cryptography
 - Consider the **symmetric** cryptography
 - Two key cryptography
 - Non-security key cryptography
- The concept is relative **newcomer**
 - In the late 1960s by **GCHQ of British**
 - Independently, in early 1970s by academic researchers

MISCONCEPTIONS ON PKC



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- **PKC is more secure than that of symm cipher**
 - Cipher Security is depends on computational work to break a cipher – both are depends on it
- **PKC made symm cipher obsolete**
 - The problem of computation overhead of PKC
- **Key distribution of PKC is trivial**
 - The procedures of PKC are so not simpler and more efficient than those of symm cipher
 - PKI is required for the key distribution of PKC

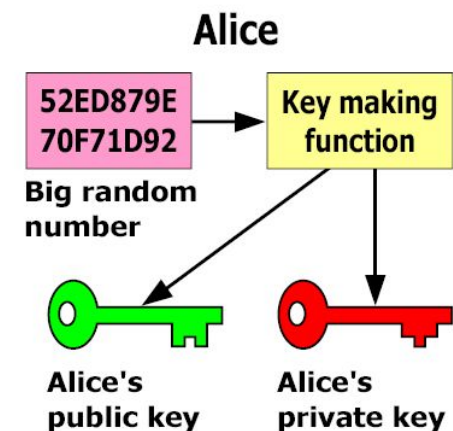
KEY GENERATION OF PKC



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- Making two keys: Based on **trap door one way function**
 - Easy to compute in one direction
 - Hard to compute in other direction
 - “Trap door” used to create keys
 - Example: Given **p** and **q**, product **$N=pq$** is easy to compute, but given **N**, it is hard to find **p** and **q**

- A message encrypted by the **public key** can be decrypted only with the corresponding **private key**

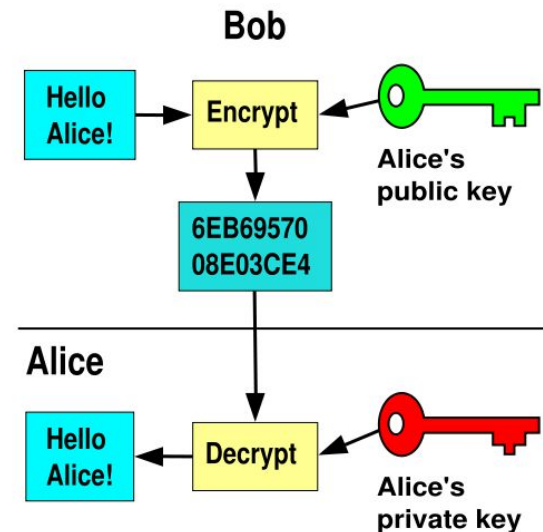


TWO MAIN BRACHES OF PKC



■ Public key **Encryption**

- Suppose we encrypt **M** with Alice's public key
- Only Alice's private key can decrypt to find **M**

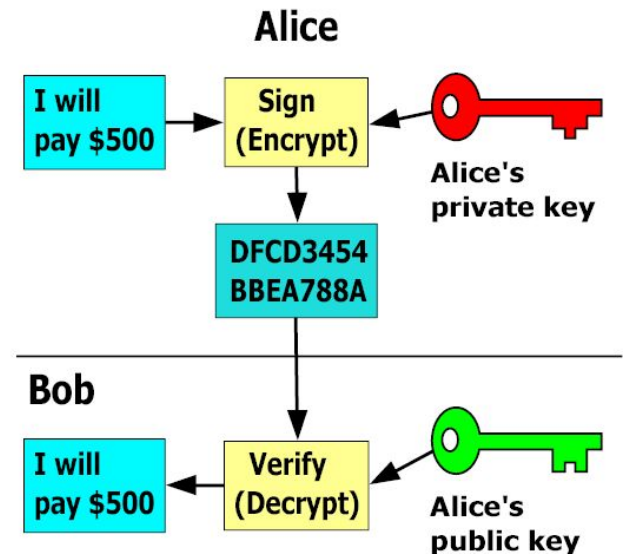


TWO MAIN BRANCHES OF PKC



■ Digital Signature

- **Sign** by “encrypting” with private key
- Anyone can **verify** signature by “decrypting” with public key
- But only private key holder could have signed
- Like a handwritten signature (and then some)



PKCS TO DISCUSS



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- **Knapnsack**
 - The first proposed PKC
 - It is inscure
- **RSA**
 - Problem of factoring large numbers
- **Diffie-Hellman Key Exchange**
 - Discrete log problem
- **ECC(Elliptic Curve Cryptography)**
 - Based on the algebraic structure of elliptic curves over finite fields

KNAPSACK



KNAPSACK PROBLEM



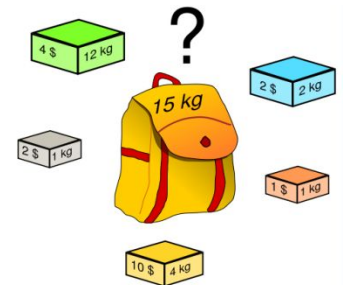
- Given a set of n weights $W_0, W_1, \dots, W_n$ and a sum S , is it possible to find $a_i \in \{0,1\}$ so that

$$S = a_0W_0 + a_1W_1, \dots + a_{n-1}W_{n-1}$$

(technically, this is “subset sum” problem)

- Example

- Weights (62,93,26,52,166,48,91,141)
- Problem: Find subset that sums to $S=302$
- Answer: $62+26+166+48=302$
- The (general) knapsack is NP-complete



KNAPSACK PROBLEM



- General knapsack (GK) is **hard** to solve
- But **superincreasing knapsack (SIK)** is easy
- **SIK** each weight greater than the sum of all previous weights
- Example
 - Weights (2,3,7,14,30,57,120,251)
 - Problem: Find subset that sums to $S=186$
 - **Work from largest to smallest weight**
 - Answer: $120+57+7+2=186$

KNAPSACK CRYPTOSYSTEM



1. Generate superincreasing knapsack (SIK)
2. Convert SIK into “general” knapsack (GK)
 - Public Key: **GK**
 - Private Key: **SIK plus conversion factors**

- Easy to encrypt with GK
- With private key, easy to decrypt (convert ciphertext to SIK)
- Without private key, must solve GK (???)

KNAPSACK CRYPTOSYSTEM



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1. Let (2,3,7,14,30,57,120,251) be the SIK
2. Choose **m** = 41 and **n** = 491
with **m**, **n** rel. prime and **n**
greater than sum of elements of
SIK

Then General knapsack can be
computed;

3. General knapsack:
(82,123,287,83,248,373,10,471)

$$2 \cdot 41 \bmod 491 = 82$$

$$3 \cdot 41 \bmod 491 = 123$$

$$7 \cdot 41 \bmod 491 = 287$$

$$14 \cdot 41 \bmod 491 = 83$$

$$30 \cdot 41 \bmod 491 = 248$$

$$57 \cdot 41 \bmod 491 = 373$$

$$120 \cdot 41 \bmod 491 = 10$$

$$252 \cdot 41 \bmod 491 = 471$$

KNAPSACK EXAMPLE



- Private key: (2,3,7,14,30,57,120,251)

$$n = 491 \quad m^{-1} \bmod n \rightarrow 41^{-1} \bmod 491 = 12$$

- Public key: (82,123,287,83,248,373,10,471)

- Example: Encrypt 10010110

$$82 + 83 + 373 + 10 = 548$$

- To decrypt

- $548 \cdot 12 = 193 \bmod 491 = S$
- Solve (easy) SIK with $S = 193$
- $193 = 2 + 14 + 57 + 120$
- Obtain plaintext 10010110

$$2 \cdot 41 \bmod 491 = 82$$

$$3 \cdot 41 \bmod 491 = 123$$

$$7 \cdot 41 \bmod 491 = 287$$

$$14 \cdot 41 \bmod 491 = 83$$

$$30 \cdot 41 \bmod 491 = 248$$

$$57 \cdot 41 \bmod 491 = 373$$

$$120 \cdot 41 \bmod 491 = 10$$

$$252 \cdot 41 \bmod 491 = 471$$

KNAPSACK WEAKNESS



- **Trapdoor:** Convert SIK into “general” knapsack using modular arithmetic
- **One-way:** General knapsack easy to encrypt, hard to solve; SIK easy to solve
- This knapsack cryptosystem is **insecure**
 - Broken in 1983 with Apple II computer
 - The attack uses **lattice reduction**
 - “General knapsack” derived from SIK is not general enough!
 - This special knapsack is easy to solve!



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RSA



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- Invented by Cocks (GCHQ), independently, by **Rivest, Shamir and Adleman** (MIT)
- Let p and q be two large prime numbers
- Let $N = pq$ be the modulus
- Choose e relatively prime to $(p-1)(q-1)$
- Find d s.t. $ed = 1 \bmod (p-1)(q-1)$
- Public key is (N, e)
- Private key is d

RSA



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- To encrypt message **M** compute
 - $C = M^e \bmod N$
- To decrypt **C** compute
 - $M = C^d \bmod N$
- Recall that **e** and **N** are public
- If attacker can factor **N**, he can use **e** to easily find **d** since $ed = 1 \bmod (p-1)(q-1)$
- **Factoring the modulus breaks RSA**
- It is not known whether factoring is the only way to break RSA



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DOES RSA REALLY WORK?

- Given $C = M^e \bmod N$ we must show

$$M = C^d \bmod N = M^{ed} \bmod N \quad \text{where } M < N$$

- **Euler's Theorem**

If M is relatively prime to N then

$$M^{\phi(N)} = 1 \bmod N \quad \text{where } \phi(N) \text{ is totient function}$$

- Facts:

- $ed = 1 \bmod (p-1)(q-1)$

- By definition of “mod”, $ed = k(p-1)(q-1) + 1$

DOES RSA REALLY WORK?



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■ Facts:

$$ed = 1 \bmod (p-1)(q-1) \quad ed = k(p-1)(q-1) + 1$$

■ By definition of “mod”,

$$■ \phi(N) = (p-1)(q-1)$$

$$■ \text{ Then } ed - 1 = k(p-1)(q-1) = k\phi(N)$$

■ Prove

$$\begin{aligned} M^{ed} &= M^{(ed-1)+1} = M \bullet M^{ed-1} = M \bullet M^{k\phi(N)} \\ &= M \bullet (M^{\phi(N)})^k \bmod N = M \bullet (1)^k \bmod N \\ &= M \bmod N \end{aligned}$$

SIMPLE RSA EXAMPLE



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- Example of RSA
 - Select “large” primes $p = 11, q = 3$
 - Then $N = pq = 33$ and $(p-1)(q-1) = 20$
 - Choose $e = 3$ (relatively prime to 20)
 - Find d such that $ed = 1 \pmod{20}$, we find that $d = 7$ works
- **Public key:** $(N, e) = (33, 3)$
- **Private key:** $d = 7$

SIMPLE RSA EXAMPLE

- **Public key:** $(N, e) = (33, 3)$

- **Private key:** $d = 7$

- Suppose message $M = 8$

- Ciphertext C is computed as

- $$C = M^e \bmod N = 8^3 = 512 = 17 \bmod 33$$

- Decrypt C to recover the message M by

$$M = C^d \bmod N = 17^7 = 410,338,673 \bmod 33$$

$$= 12,434,505 \times 33 + 8 = 8 \bmod 33$$

MORE EFFICIENT RSA (I)



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- Modular exponentiation example
 - $5^{20} = 95367431640625 = 25 \text{ mod } 35$
- A better way: **repeated squaring**
 - $20 = 10100 \text{ base } 2$
 - $(1, 10, 101, 1010, 10100) = (1, 2, 5, 10, 20)$
 - Note that $2 = 1 \cdot 2$, $5 = 2 \cdot 2 + 1$, $10 = 2 \cdot 5$, $20 = 2 \cdot 10$
 - $5^1 = 5 \text{ mod } 35$
 - $5^2 = (5^1)^2 = 5^2 = 25 \text{ mod } 35$
 - $5^5 = (5^2)^2 \cdot 5^1 = 25^2 \cdot 5 = 3125 = 10 \text{ mod } 35$
 - $5^{10} = (5^5)^2 = 10^2 = 100 = 30 \text{ mod } 35$
 - $5^{20} = (5^{10})^2 = 30^2 = 900 = 25 \text{ mod } 35$
- **Never have to deal with huge numbers!**

MORE EFFICIENT RSA (2)



- Let $e = 3$ for all users (but not same N or d)
 - Public key operations only require 2 multiplies
 - Private key operations remain “expensive”
 - If $M < N^{1/3}$ then $C = M^e = M^3$ and **cube root attack**
 - (mod N) operation has no effect
 - For any M , if C_1, C_2, C_3 sent to 3 users, cube root attack works (**uses Chinese Remainder Theorem**)
 - Can prevent cube root attack by padding message with random bits
- Note: $e = 2^{16} + 1$ also used: Protect CRT attack



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- GALOIS FIELD
Check book
- Understanding cryptography Page 98